

# LAB3\_Report

MEDICAL IMAGE AND SIGNAL PROCESSING LAB – SPRING 2026  
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# 1. Data Description

## (a) Time – Domain Representation

Using the MATLAB script below, we plotted the wanted ECG signal:

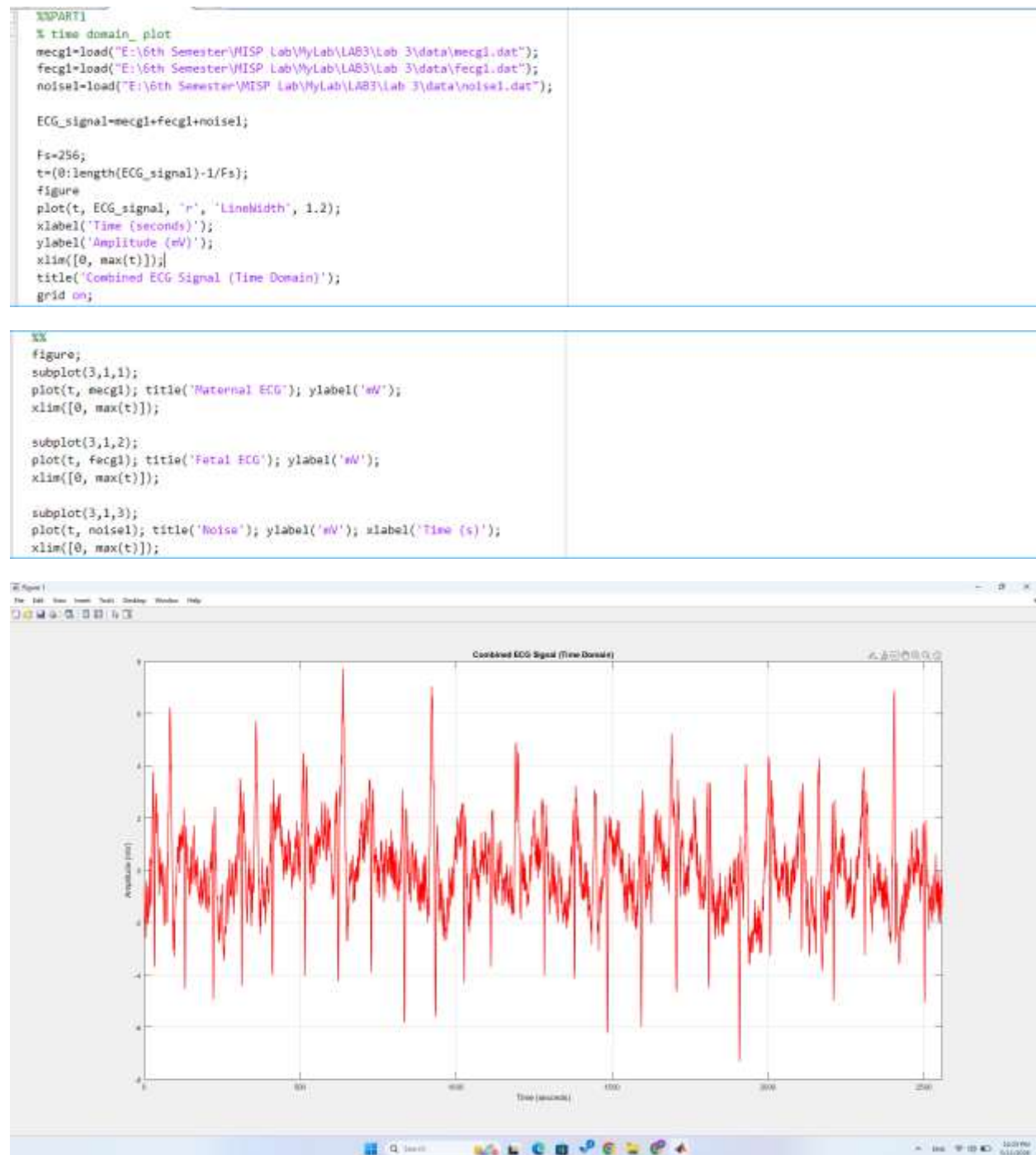


Figure1: Mixed ECG Signal Time – Domain Representation

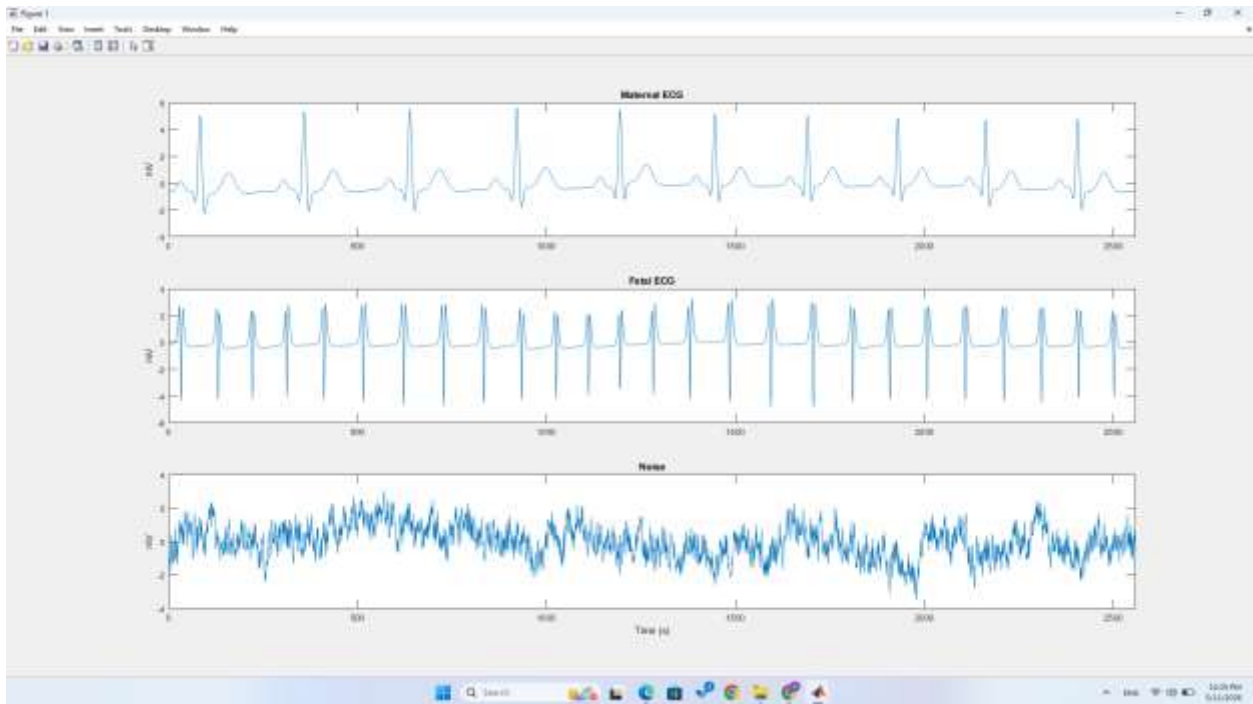


Figure 2: Mother, Fetal and Noise Signals Time – Domain Representation

### (b) Power Spectrum Analysis

Using the MATLAB script below, we represent the power spectrum for the ECG signals from Mother, Fetal and the Noise:

```

31 % PART 2
32 fs = 256;
33
34 [pwx_m, f_m] = pwelch(mecg1, [], [], [], fs);
35 [pwx_f, f_f] = pwelch(fecg1, [], [], [], fs);
36 [pwx_n, f_n] = pwelch(noisal, [], [], [], fs);
37
38 figure;
39
40 subplot(3,1,1)
41 plot(f_m, 10*log10(pwx_m))
42 title('Maternal ECG Power Spectrum')
43 xlabel('Frequency (Hz)')
44 ylabel('Power (dB/Hz)')
45 xlim([0 80])
46 grid on
47
48 subplot(3,1,2)
49 plot(f_f, 10*log10(pwx_f))
50 title('Fetal ECG Power Spectrum')
51 xlabel('Frequency (Hz)')
52 ylabel('Power (dB/Hz)')
53 xlim([0 80])
54 grid on
55
56 subplot(3,1,3)
57 plot(f_n, 10*log10(pwx_n))
58 title('Noise Power Spectrum')
59 xlabel('Frequency (Hz)')
60 ylabel('Power (dB/Hz)')
61 xlim([0 80])
62 grid on
63

```

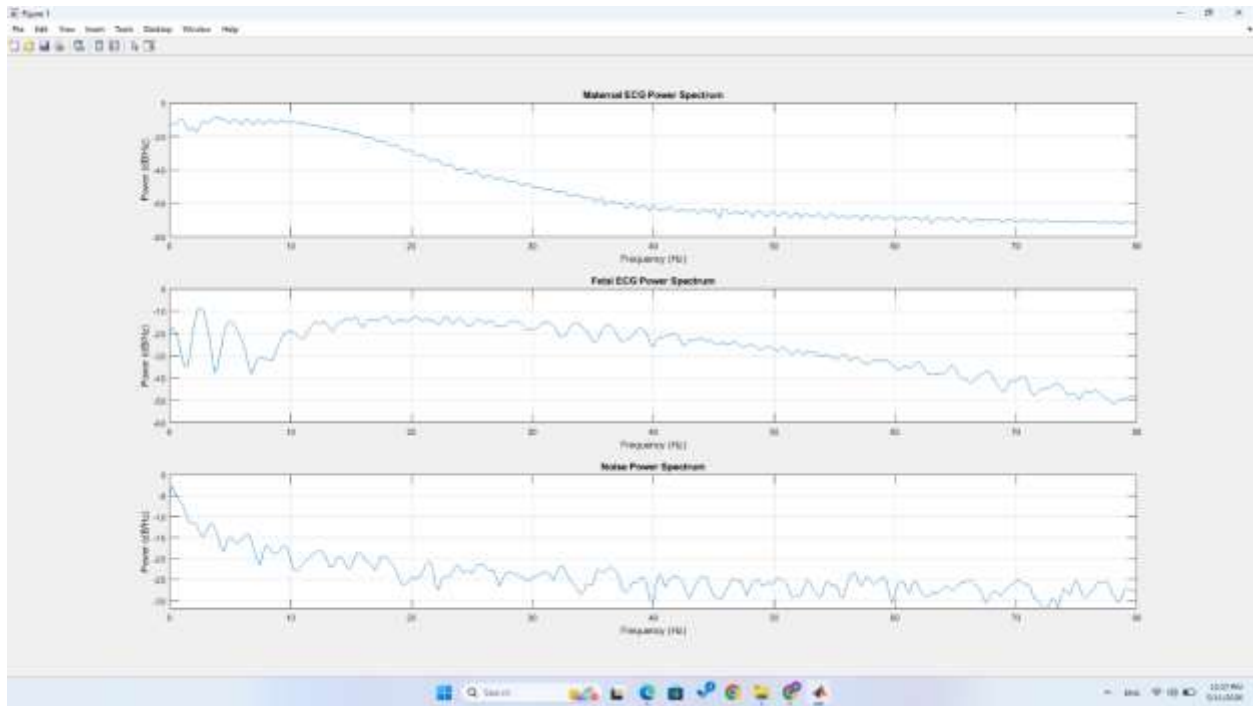


Figure 3: Power Spectrum of Mother, Fetal and Noise Signals

The power spectra of maternal and fetal ECGs are similar in overall trend and slope, and both signals lack components above 100 Hz. Despite these similarities, their power distribution differs: maternal ECG power drops off sharply at 20 Hz, whereas fetal ECG power persists until 70 Hz. This demonstrates that the fetal ECG contains higher frequency information than its maternal counterpart.

### (c) Mean and Variance Analysis and (d) Histogram Analysis

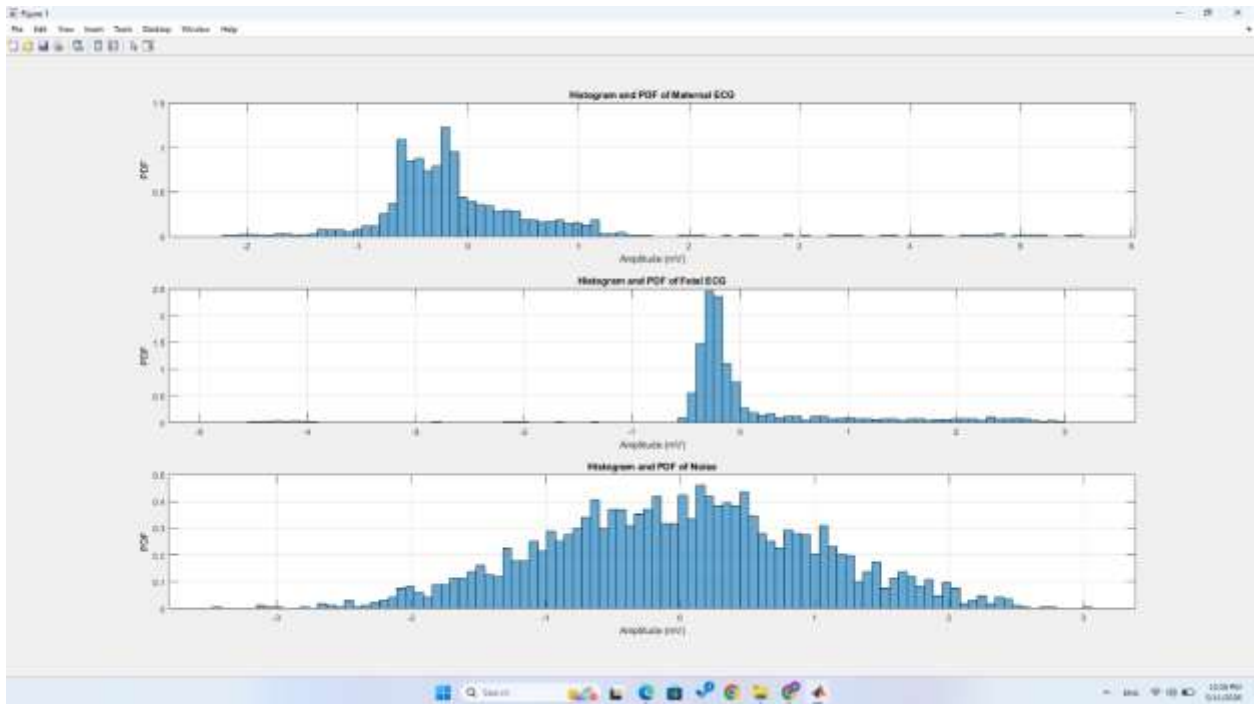
Using the MATLAB script below, we examined the mean and variance of the three wanted signals, and also plotted their histogram:

```
31 % PART 2
32 fs = 256;
33
34 [pwx_m, f_m] = pwelch(mecg1, [], [], [], fs);
35 [pwx_f, f_f] = pwelch(fecg1, [], [], [], fs);
36 [pwx_n, f_n] = pwelch(noise1, [], [], [], fs);
37
38 figure;
39
40 subplot(3,1,1)
41 plot(f_m,10*log10(pwx_m))
42 title('Maternal ECG Power Spectrum')
43 xlabel('Frequency (Hz)')
44 ylabel('Power (dB/Hz)')
45 xlim([0 80])
46 grid on
47
48 subplot(3,1,2)
49 plot(f_f,10*log10(pwx_f))
50 title('Fetal ECG Power Spectrum')
51 xlabel('Frequency (Hz)')
52 ylabel('Power (dB/Hz)')
53 xlim([0 80])
54 grid on
55
56 subplot(3,1,3)
57 plot(f_n,10*log10(pwx_n))
58 title('Noise Power Spectrum')
59 xlabel('Frequency (Hz)')
60 ylabel('Power (dB/Hz)')
61 xlim([0 80])
62 grid on
63
```



```
Command Window
Maternal ECG: Mean = -0.0000 | Variance = 1.0000
Fetal ECG: Mean = -0.0000 | Variance = 1.0000
Noise: Mean = -0.0000 | Variance = 1.0000
Kurtosis (Maternal ECG): 14.0421
Kurtosis (Fetal ECG): 8.9801
Kurtosis (Noise): 2.7662
M >>
```

Based on the elicited results, we can confirm that both the variance and the mean of these signals are equal so the variance of the signals has no relevance with their frequency content and we can't use these two parameters to compare signals or differentiate them.



The values of the kurtosis were demonstrated in the previous figure along with the mean and variance. Thus, from both the Histograms and the Kurtosis values we can suggest the lesser the kurtosis value is, the more similar the signal's pdf gets to a Gaussian PDF.

## 2. Signal Separation Using SVD

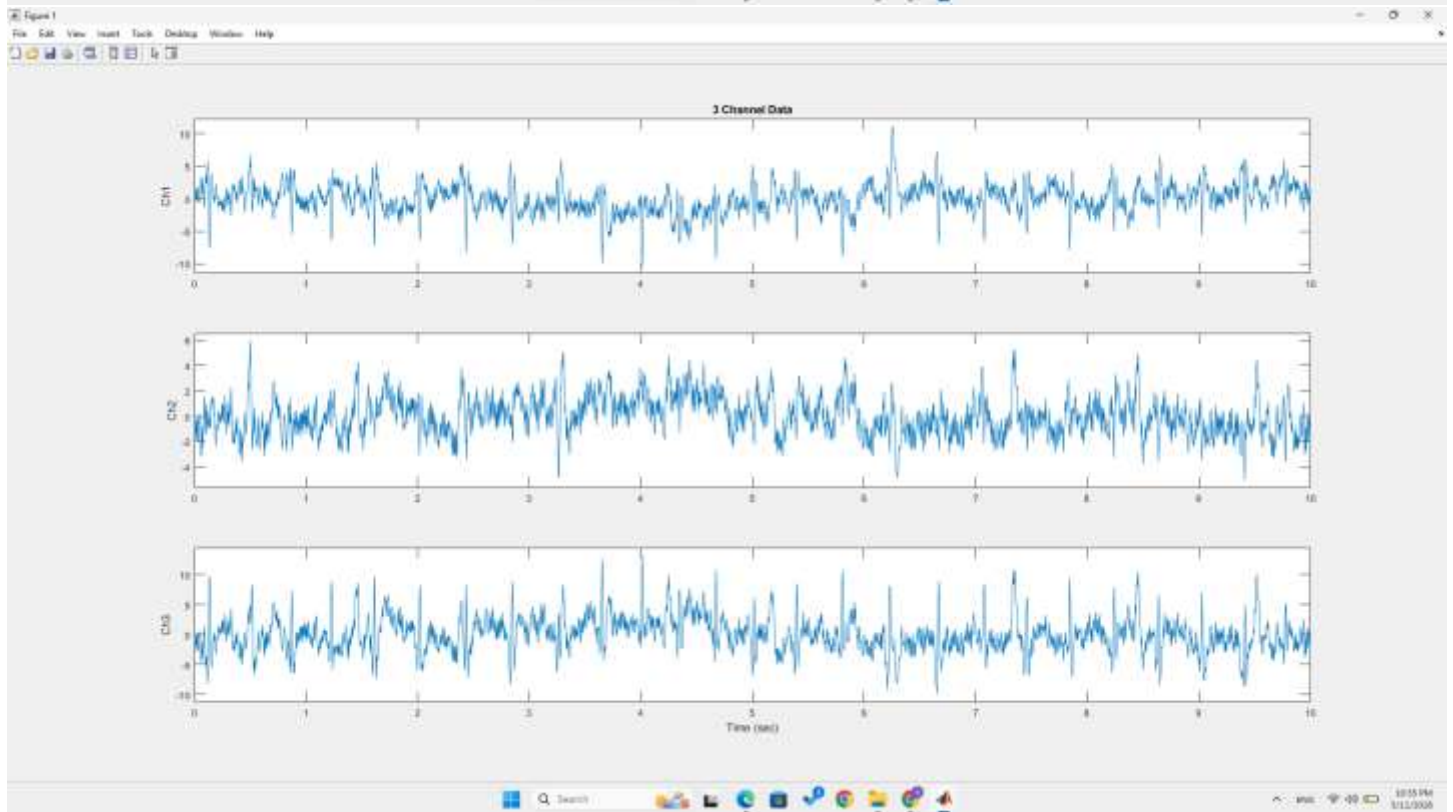
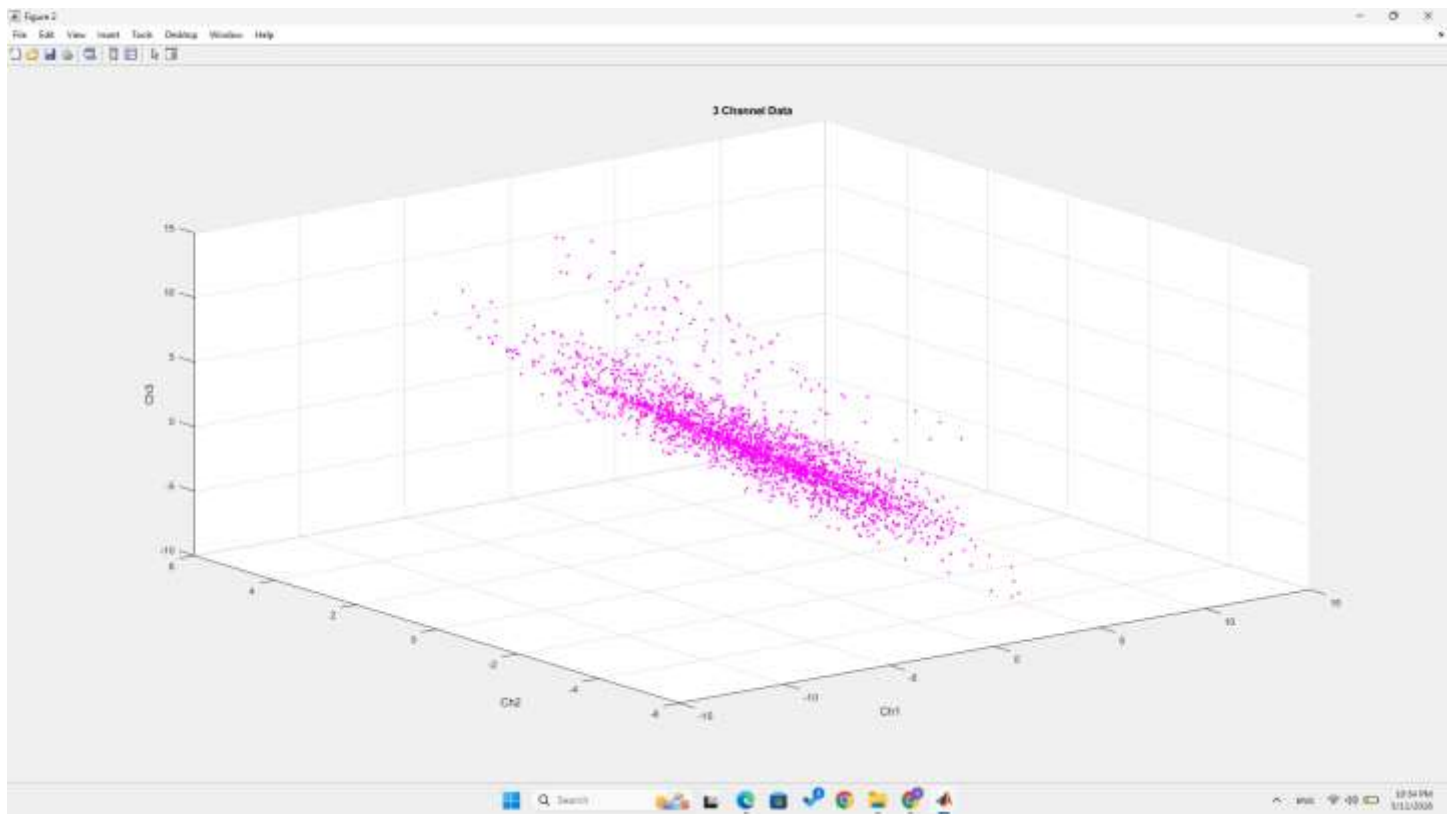
(a) Representation of  $X$  data and  $S$ ,  $U$  and  $V$  Extraction

Using the MATLAB code below, we obtain the required results:

```

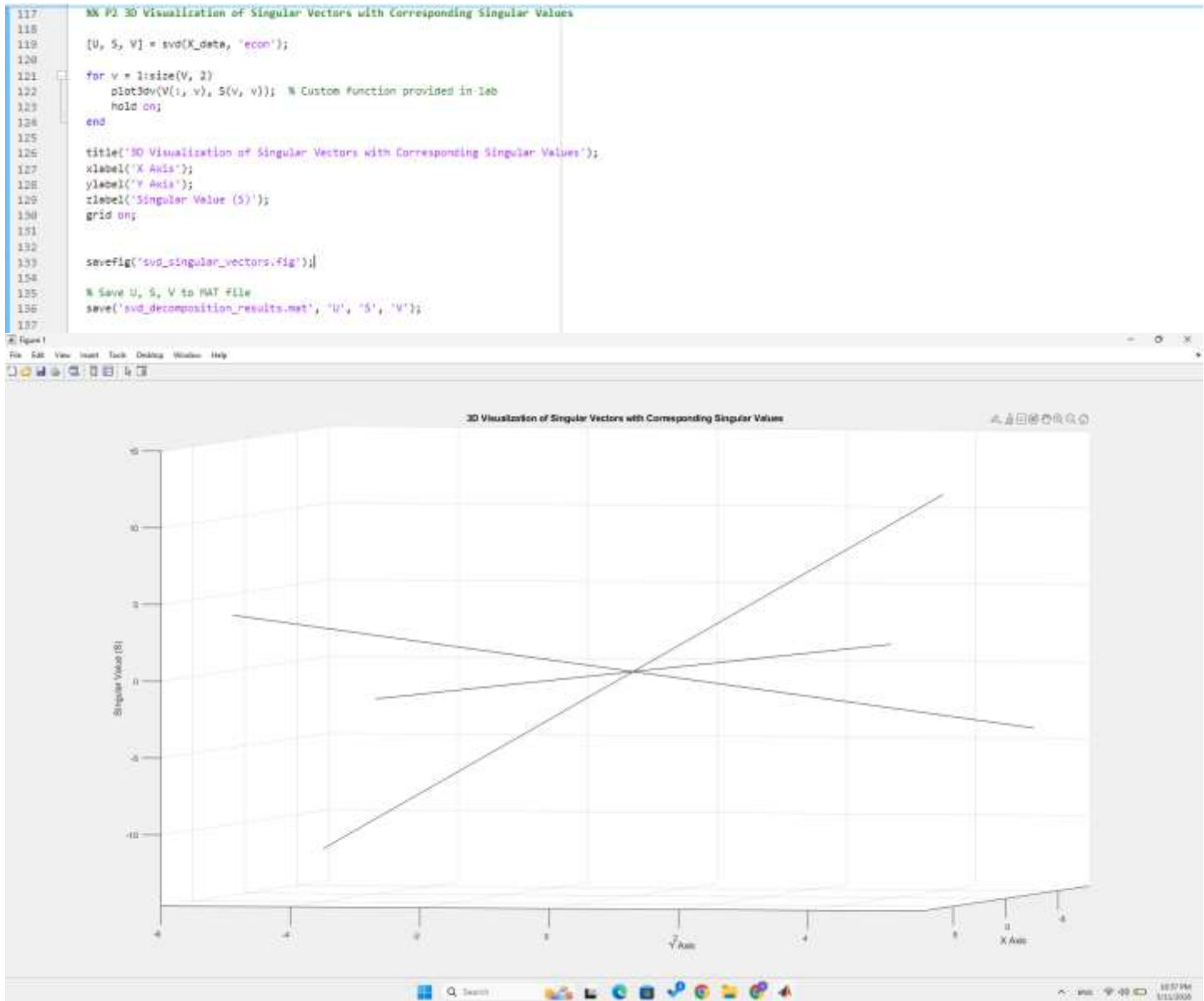
111 %%% SVD PART2
112 %%%
113 X_data=load('E:\6th Semester\MEIS Lab\MyLab\LAB 3\data\X.mat');
114 addpath('E:\6th Semester\MEIS Lab\MyLab\LAB 3\matlab');
115 plot3ch(X_data)
116

```



## (b) Plotting Columns of $V$ Matrix and Saving the Results

Using the MATLAB script below, we plotted the columns of  $V\_MAT$  and also saved our results for further investigation.

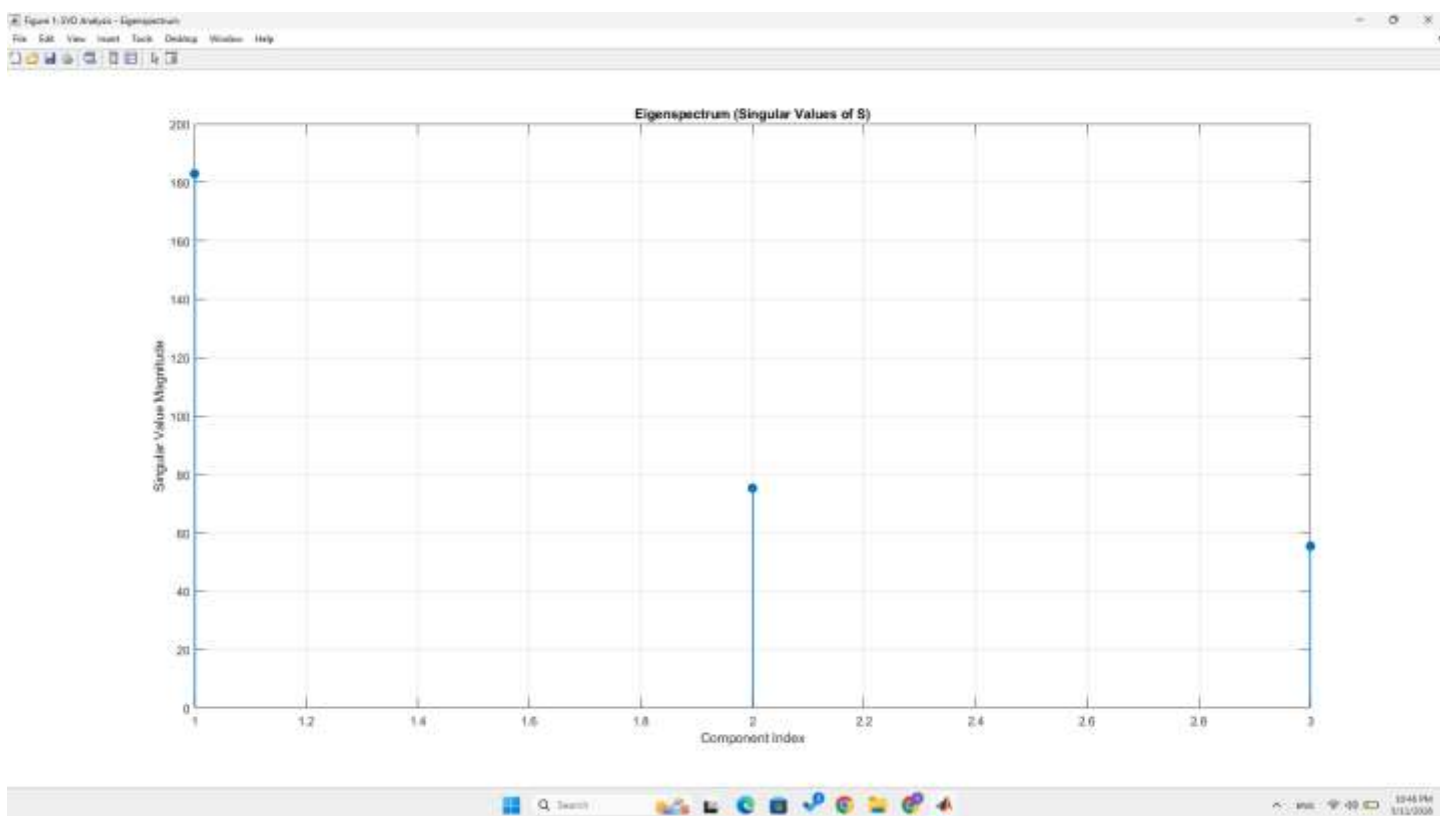
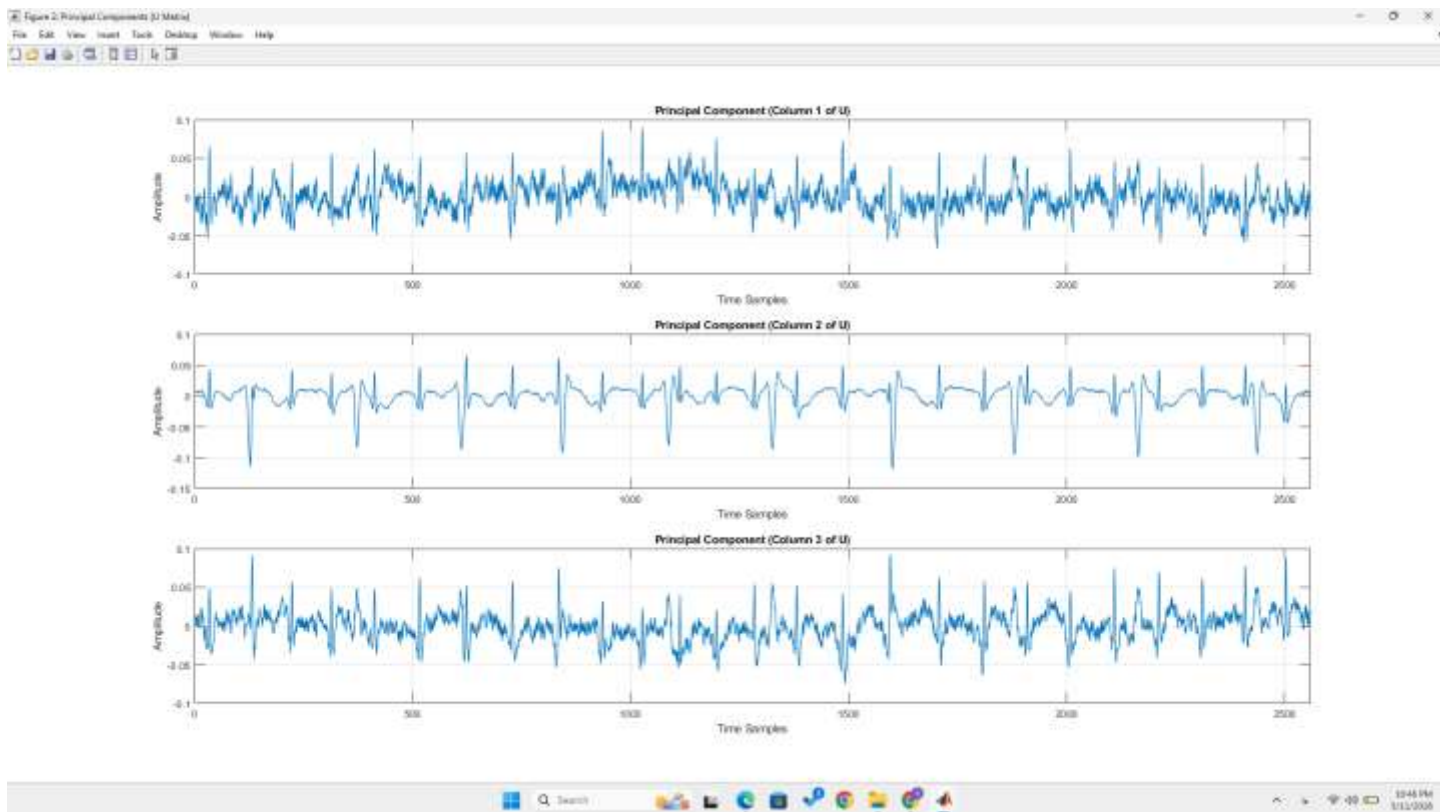




### (c) *U Matrix Analysis*

Using the MATLAB script below, we plotted the required results. As we can observe down below, the Subplots 1 and 3 contain a considerable amount of noise, but the subplot2 seems to be cleaner, with a mixture of mother and fetal ECG, Also the third channel is more similar to that of the mother ECG. Regarding the fact that fetal ECG includes sharper, negative peaks compared to that of the mother ECG, this component is a suitable candidate to be kept for reconstruction. In order to keep the data of the fetal, we kept the singular value related to the fetal, and ignored other values.

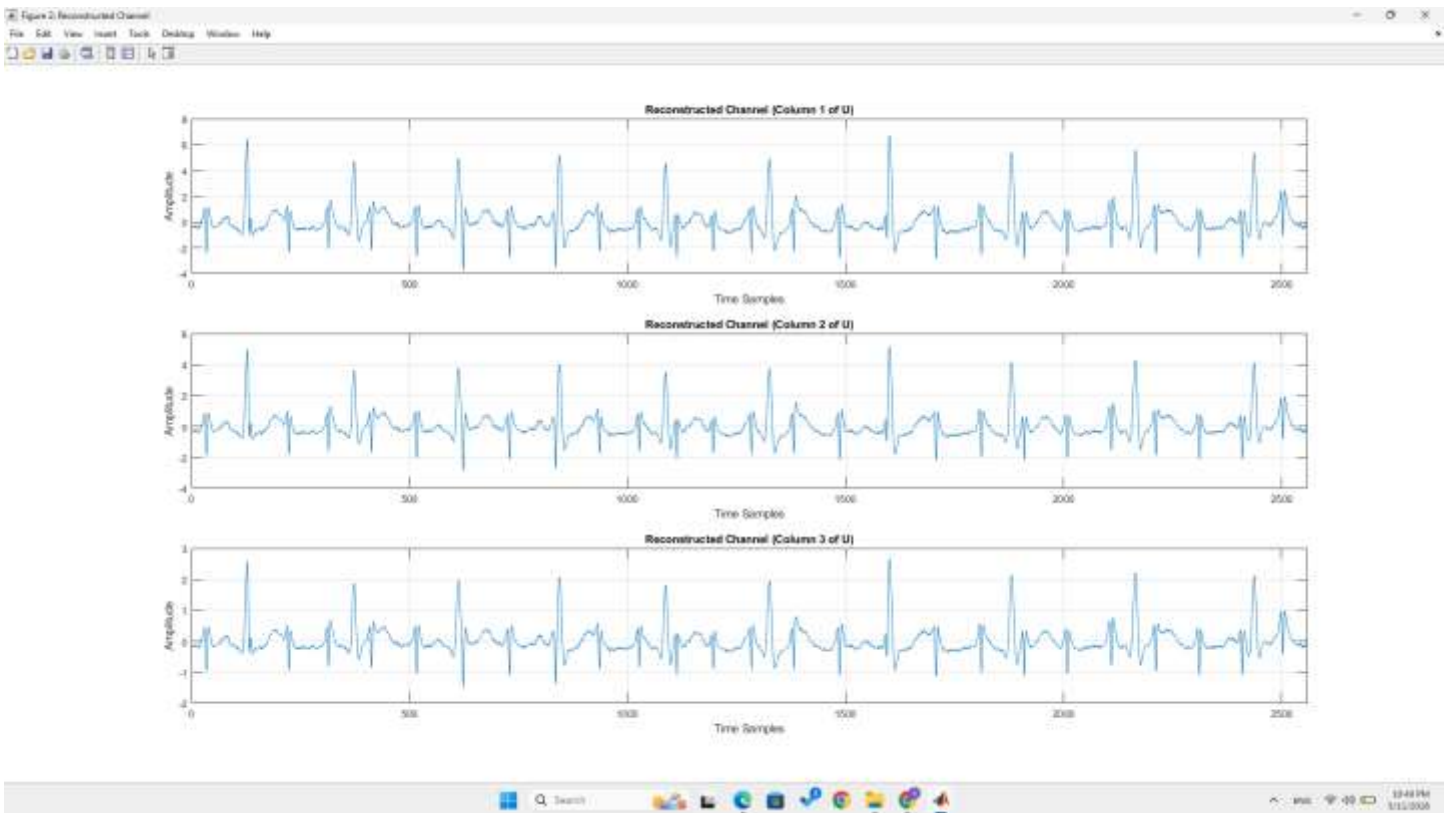
```
138 %%% p3
139
140
141 % 3. Extract Singular Values (Diagonal elements of S)
142 singular_values = diag(S);
143
144 % 4. Plotting the Eigenspectrum
145 figure('Name', 'SVD Analysis - Eigenspectrum', 'Color', 'w');
146 stem(singular_values, 'filled', 'MarkerSize', 8, 'LineWidth', 1.5);
147 grid on;
148 title('Eigenspectrum (Singular Values of S)');
149 xlabel('Component Index');
150 ylabel('Singular Value Magnitude');
151 set(gcf, 'FontSize', 12);
152
153 % 5. Plotting the Principal Components (Columns of U)
154 figure('Name', 'Principal Components (U Matrix)', 'Color', 'w');
155
156 for i = 1:3
157     subplot(3, 1, i);
158     plot(U(:, i), 'linewidth', 1);
159     title(['Principal Component (Column ', num2str(i), ' of U)']);
160     xlabel('Time Samples');
161     ylabel('Amplitude');
162     xlim([0, max(t)]);
163     grid on;
164 end
165 fprintf('Size of U: %d x %d\n', size(U,1), size(U,2));
166 fprintf('Size of S: %d x %d\n', size(S,1), size(S,2));
167 fprintf('Size of V: %d x %d\n', size(V,1), size(V,2));
```



### (d) Reconstruction of Fetal ECG

Using the MATLAB code below, we reconstructed the signal:

```
205 % plot
206 S_new = S;
207 S_new(:, 1) = 0;
208 S_new(:, 3) = 0;
209 X_reconstructed = U * S_new * V';
210 figure('Name', 'Reconstructed Channel', 'Color', 'w');
211
212 for i = 1:3
213     subplot(3, 1, i);
214     plot(X_reconstructed(:, i), 'linewidth', 0.5);
215     title(['Reconstructed Channel (Column ', num2str(i), ' of U)']);
216     xlabel('Time Samples');
217     ylabel('Amplitude');
218     xlim([0, max(t)]);
219     grid on;
220 end
221
222
```



It is clear that after applying SVD, the fetal ECG was reconstructed successfully, suppressing noise signal. However, it should be noted that there is still mother ECG in the signal, suggesting that this approach cannot effectively isolate fetal ECG.

### 3. Source Separation Using ICA

#### (a) *Applying ICA on Data*

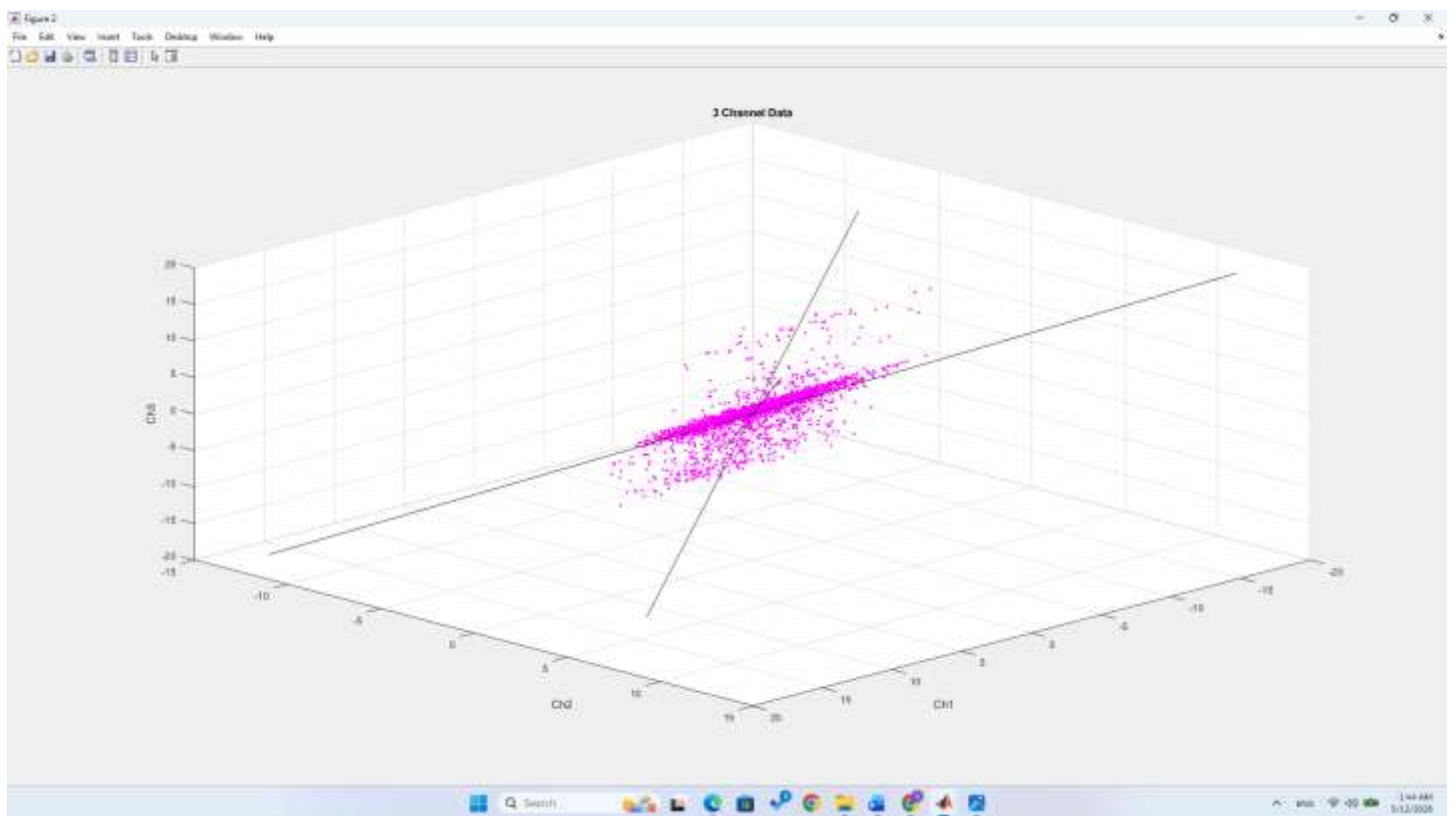
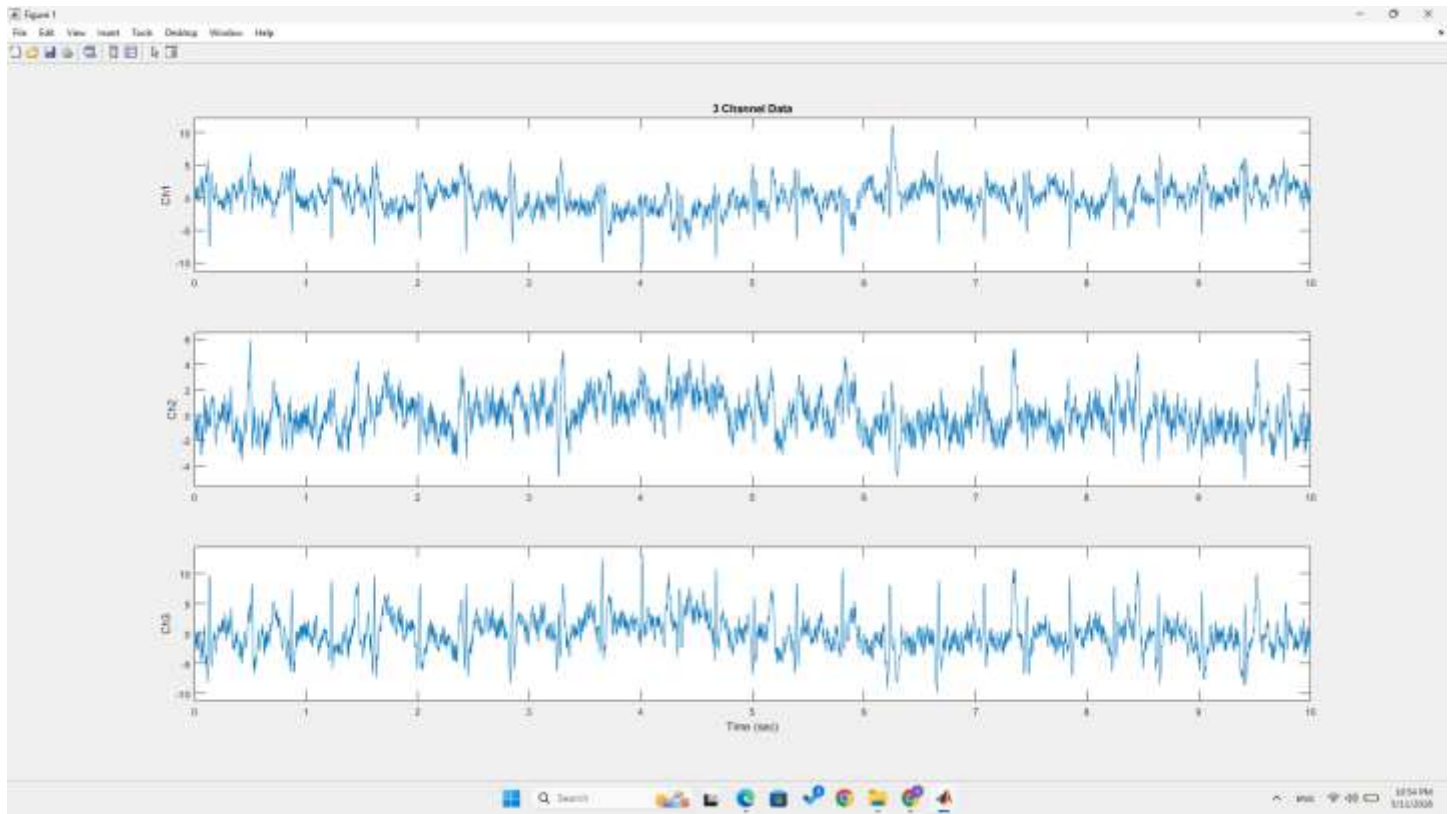
In this part, we applied the ICA algorithm using the MATLAB script below and saved proper matrices.

```
%% Part 3.1
addpath('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\data')
addpath('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\matlab')
X_mat = load('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\data\X.dat');
X_mat_t = X_mat';
[H_mat, Z_mat] = ica(X_mat_t);
A = inv(H_mat);
save('Z_mat.mat', 'Z_mat');
save('H_mat.mat', 'H_mat');
save('A_mat.mat', 'A');
```

#### (b) *Scatter Plot and Component Representation*

Using the MATLAB script below, we plotted the scatter plots and visualized the components.

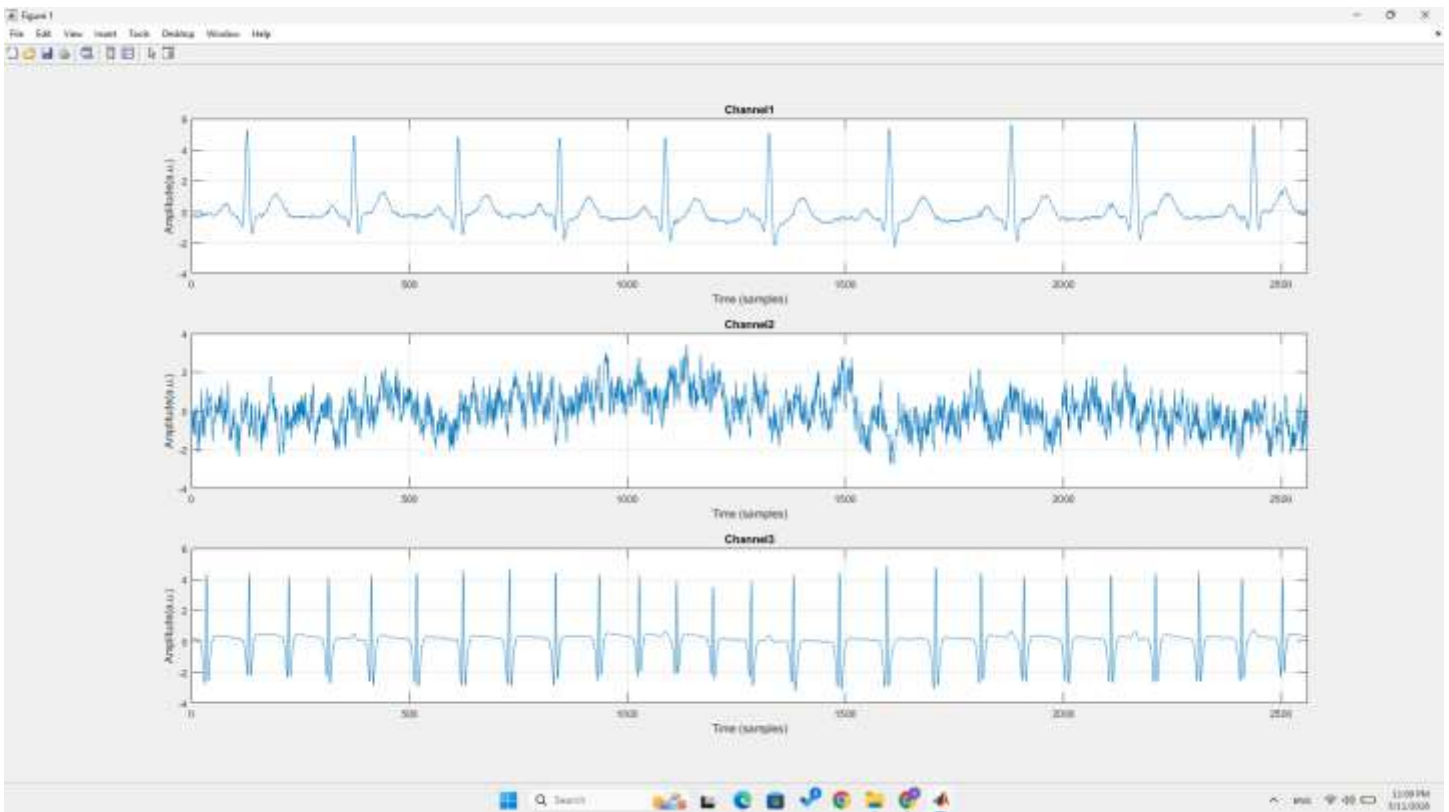
```
4 %% Part 3.1
5 addpath('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\data')
6 addpath('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\matlab')
7 X_mat = load('E:\6th Semester\MISP Lab\MyLab\LAB3\Lab 3\data\X.dat');
8 X_mat_t = X_mat';
9 [H_mat, Z_mat] = ica(X_mat_t);
10 A = inv(H_mat);
11 save('Z_mat.mat', 'Z_mat');
12 save('H_mat.mat', 'H_mat');
13 save('A_mat.mat', 'A');
14 %% Part 3.2.1
15 c1c;
16 plot3ch(X_mat);
17 %% Part 3.2.2
18 c1c;
19 plot3dv(A(:,1));
20 plot3dv(A(:,2));
21 plot3dv(A(:,3));
22 savefig(gcf,'part3.fig');
```



### (c) *Z\_Mat Analysis*

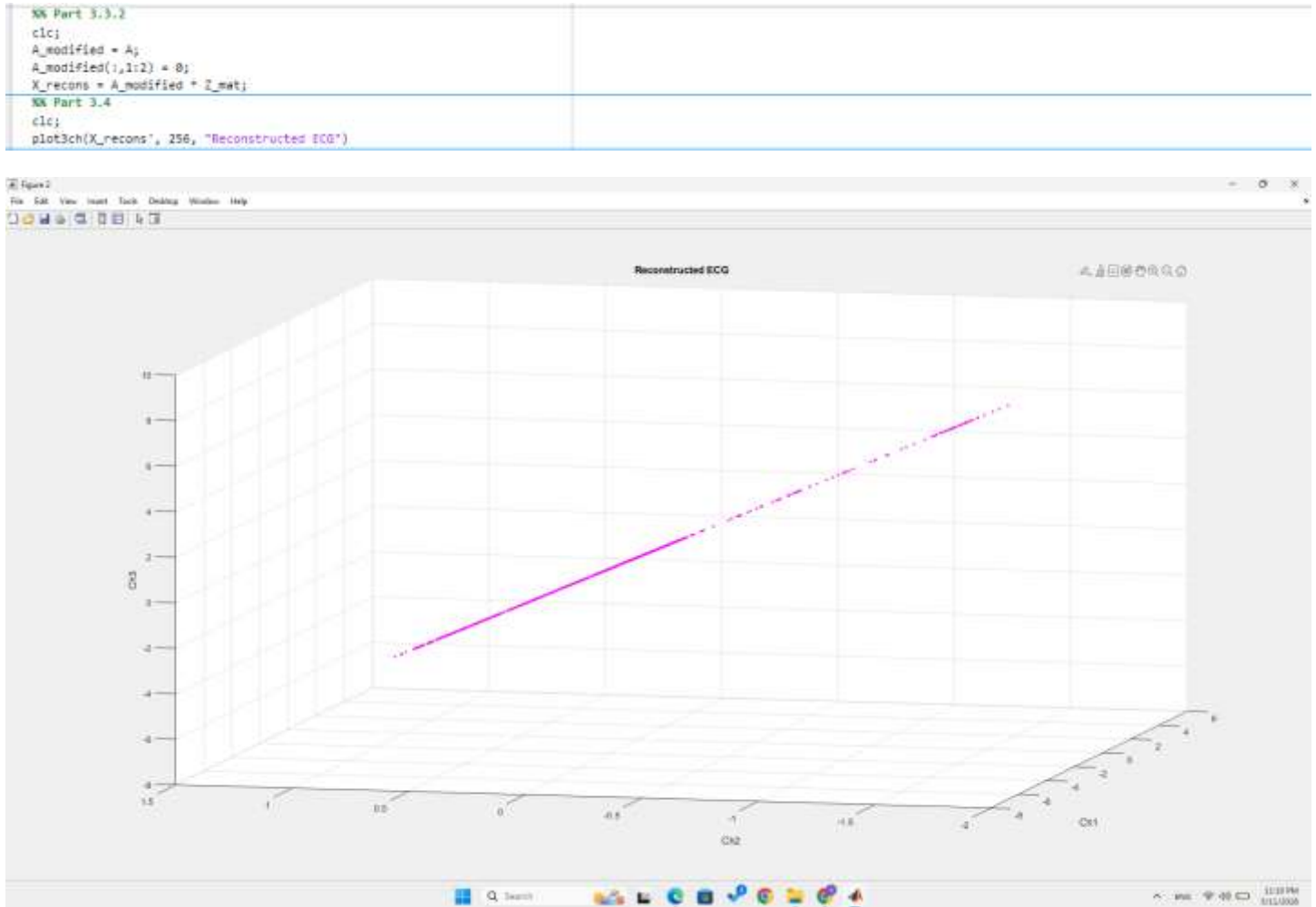
Using the MATLAB script below, we plotted the columns of  $Z$ . We can observe through the plots that the third component has the highest correlation with the fetal ECG, while components 1 and 2 are respectively related to the mother ECG and noise Signal, So we kept the 3<sup>rd</sup> component and ignored the others for the reconstruction stage.

```
23 % Part 3.3.1
24 clc;
25 figure;
26 for i = 1:3
27     subplot(3,1,i);
28     plot(Z_mat(1,i));
29     title(['Channel', num2str(i)]);
30     xlabel('Time (samples)');
31     ylabel('Amplitude(a.u.)');
32     xlim([0 2560]);
33     grid on;
34 end
```

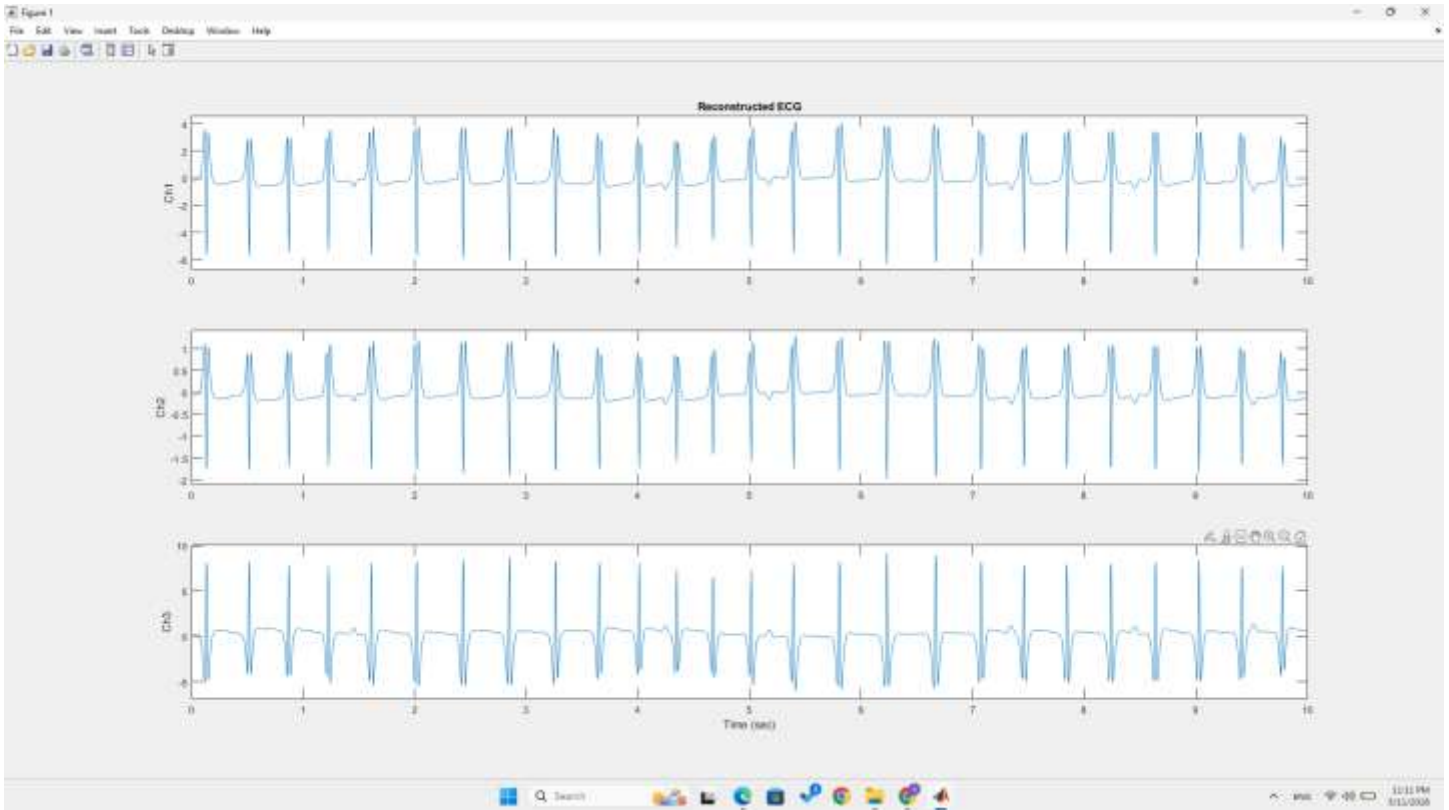


#### (d) *Fetal ECG Reconstruction*

Using the MATLAB code, and with the descriptions from the last part, we reconstruct the signal. The results suggest fine reconstruction of the fetal ECG signal from the mixed signal.







## 4. Results Comparison

### (a) Visual Comparison

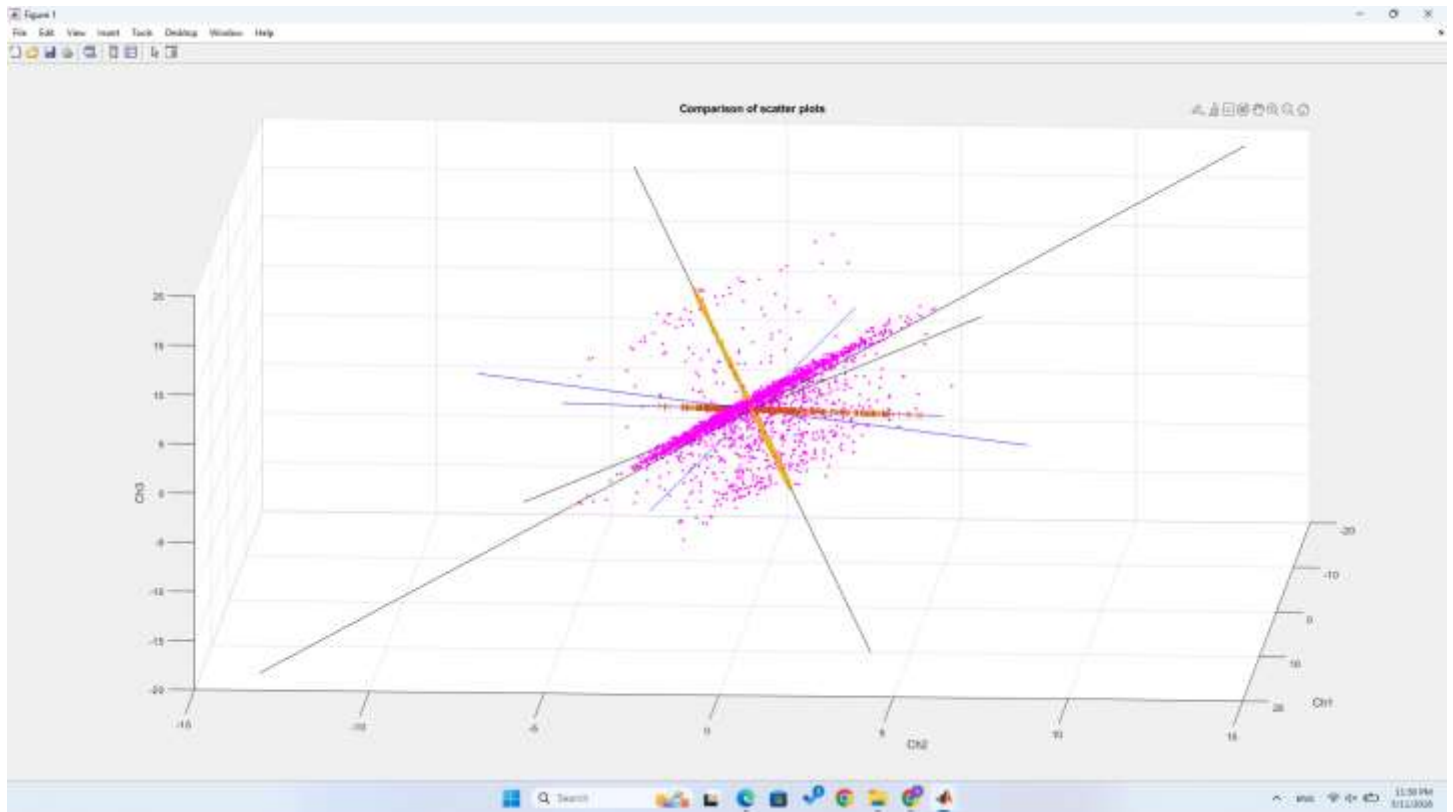
Using the MATLAB script below, we plotted the required plots

```
%% Part 4.1
figure;
plot3(X_mat(:,1), X_mat(:,2), X_mat(:,3), 'b');
hold all;
plot3(X_reconstructed(:, 1), X_reconstructed(:, 2), X_reconstructed(:, 3), 'r'); % This is from SVD
hold all;
plot3(X_recons(:, 1), X_recons(:, 2), X_recons(:, 3), 'g'); % This is from ICA
xlabel('Ch1');
ylabel('Ch2');
zlabel('Ch3');
title('Comparison of scatter plots');
grid on;
plot3dv(A(:,1), 100, 'black');
plot3dv(A(:,2), 100, 'black');
plot3dv(A(:,3), 100, 'black');
plot3dv(V(:,1), 100, 'blue');
plot3dv(V(:,2), 100, 'blue');
plot3dv(V(:,3), 100, 'blue');

norm_A = [norm(A(:,1)), norm(A(:,2)), norm(A(:,3))];
angles_A = [dot(A(:,1), A(:,2))/norm_A(1)/norm_A(2), dot(A(:,1), A(:,3))/norm_A(1)/norm_A(3), dot(A(:,2), A(:,3))/norm_A(2)/norm_A(3)];
angles_A = acosd(angles_A);

norm_V = [norm(V(:,1)), norm(V(:,2)), norm(V(:,3))];
angles_V = [dot(V(:,1), V(:,2))/norm_V(1)/norm_V(2), dot(V(:,1), V(:,3))/norm_V(1)/norm_V(3), dot(V(:,2), V(:,3))/norm_V(2)/norm_V(3)];
angles_V = acosd(angles_V);
```





Command Window

```
>> angles_A
angles_A =
    62.0287    71.3380    37.6082

>> norm_A
norm_A =
    1.6380    2.9138    2.9079

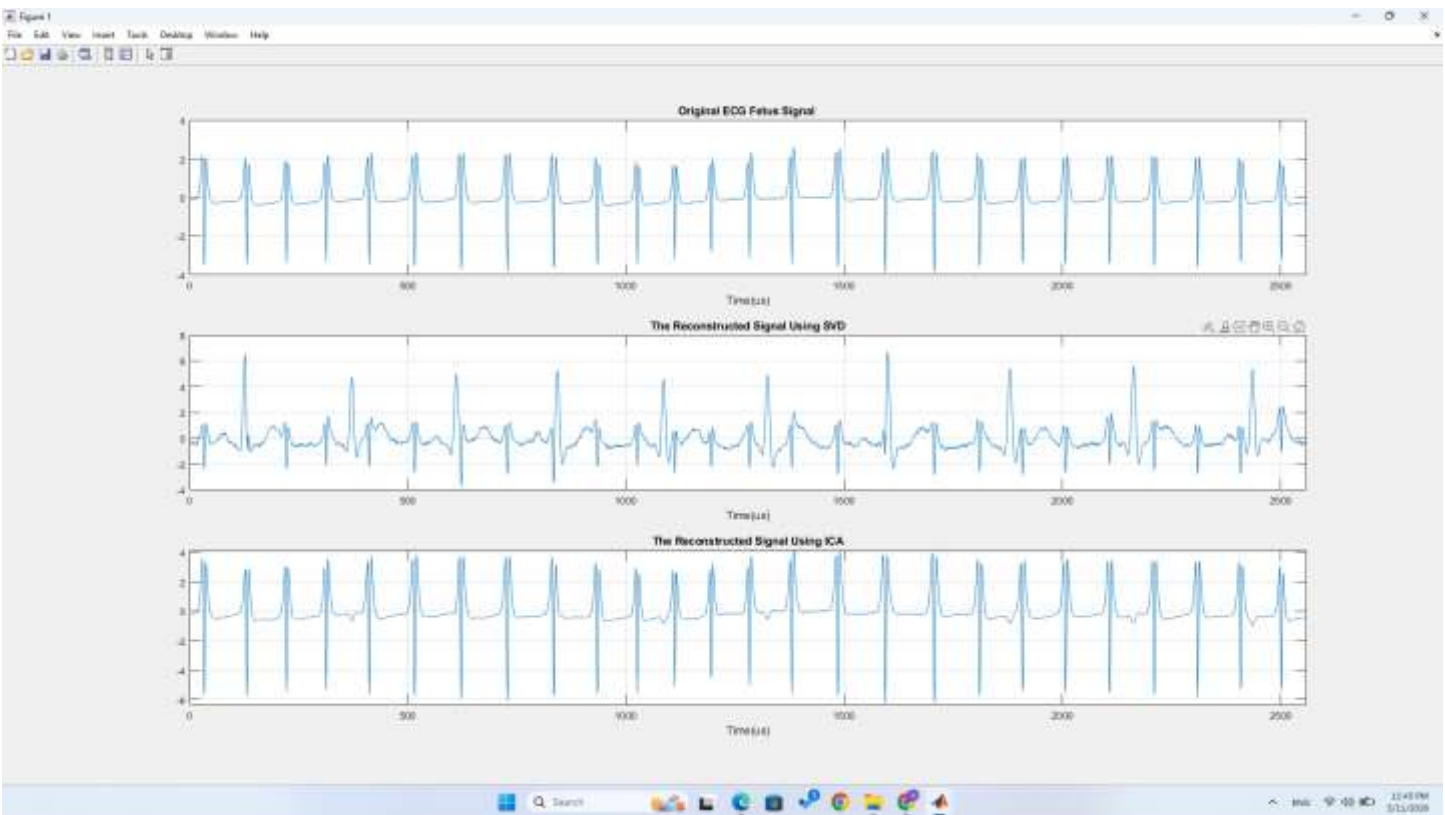
>> angles_V
angles_V =
    90.0000    90.0000    90.0000

>> norm_V
norm_V =
    1.0000    1.0000    1.0000

fx >> |
```

## (b) Comparison Using the Ideal Signal

```
292 %X Part 4.2
293 c1c;
294 FECG2 = load('E:\6th Semester\MEISP Lab\MyLab\LAB3\Lab 3\data\fecg2.dat');
295 figure;
296 subplot(3,1,1);
297 plot(t, FECG2);
298 hold on;
299 title('Original ECG Fetus Signal');
300 xlabel('Time(us)');
301 xlim([0 max(t)]);
302 grid on;
303
304 subplot(3,1,2);
305 plot(t, X_reconstructed(t,1));
306 hold on;
307 title('The Reconstructed Signal Using SVD');
308 xlabel('Time(us)');
309 xlim([0 max(t)]);
310 grid on;
311
312 subplot(3,1,3);
313 plot(t, X_recons{1, :});
314 hold on;
315 title('The Reconstructed Signal Using ICA');
316 xlabel('Time(us)');
317 xlim([0 max(t)]);
318 grid on;
```



Comparing the two reconstructed signals, we can suggest that ICA has done a significantly better job at reconstructing the fetal ECG.

### (c) Correlation Coefficient Calculation



```
319 % 4.4
320 c1c1
321 corr_svd = corrcoef(X_reconstructed(:,1), FECG2(:,1));
322 corr_ica = corrcoef(X_recons(:,1), FECG2(:,1));
323
324
325
326
327
```

```
>> corr_svd
corr_svd =
    1.0000    0.4955
    0.4955    1.0000

>> corr_ica
corr_ica =
    1.0000    0.9978
    0.9978    1.0000
```

From the correlation coefficient's calculation, we can confirm our previous findings regarding the better performance of ICA. The correlation coefficient in ICA is 0.9978 which is much higher than 0.4955 which belongs to SVD.

### (d) Comparison of ICA and SVD

Independent Component Analysis (ICA) and Singular Value Decomposition (SVD) are both widely used techniques for extracting fetal ECG from abdominal recordings that contain maternal ECG and noise, but they differ significantly in their assumptions and performance. ICA separates signals by assuming that the underlying sources, such as maternal ECG, fetal ECG, and noise, are statistically independent. Because of this, ICA is often very effective at isolating fetal ECG components and preserving waveform details, especially when multiple recording channels are available. However, its performance depends strongly on the validity of the independence assumption, and it may fail when the signals overlap significantly or when only a small number of channels are used. In addition, ICA can be computationally more complex and sensitive to noise.

On the other hand, SVD is based on matrix decomposition and separates signal components according to their energy or variance rather than statistical independence. It is simpler and more stable in noisy environments, making it useful for reducing noise and removing dominant maternal ECG components. SVD also works reasonably well even when the independence assumption is not satisfied. However, because fetal ECG usually has much lower energy than maternal ECG, important fetal components may sometimes be suppressed along with noise, leading to partial loss of signal details. Therefore, ICA generally provides better source separation and fetal ECG reconstruction quality, while SVD offers better robustness and simplicity for denoising applications.

*(e) Conclusion*

We can eventually conclude from the explanations of part (d) and from the elicited results that ICA is much more suitable for source separation than SVD for reconstructing the fetal ECG and SVD is more suitable for denoising and dimensionality reduction.